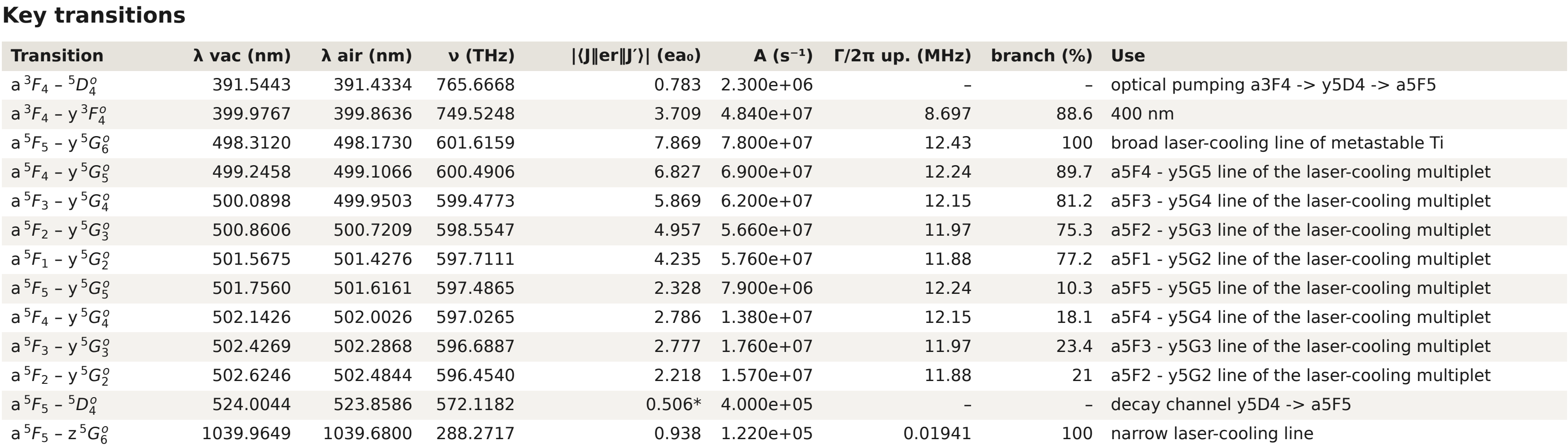


Ti<sup>+</sup> ionisation limit 55072.50 cm<sup>-1</sup> = 6.82812 eV (181.579 nm)

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000



$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available.  $A$ : measured value or NIST.  $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

| Level      | A (MHz)      | B (MHz)    | Shift of each F level from the centre of gravity (MHz) |                 |                |                 |                 |                  |  |  |
|------------|--------------|------------|--------------------------------------------------------|-----------------|----------------|-----------------|-----------------|------------------|--|--|
| $a^3F_4$   | -52.9047(7)  | 28.082(9)  | F=1/2: +545.90                                         | F=3/2: +458.96  | F=5/2: +316.87 | F=7/2: +213.84  | F=9/2: -114.23  | F=11/2: -389.76  |  |  |
| $a^3F_6$   | -37.9918(6)  | 37.681(10) | F=3/2: +493.40                                         | F=5/2: +387.32  | F=7/2: +243.04 | F=9/2: +64.81   | F=11/2: -141.92 | F=13/2: -370.50  |  |  |
| $a^3F_8$   | -74.8450(25) | -26.67(5)  | F=5/2: +1110.25                                        | F=7/2: +856.99  | F=9/2: +527.93 | F=11/2: +120.49 | F=13/2: -368.49 | F=15/2: -942.73  |  |  |
| $z^3G_5^o$ | -85(9)       | 77(5)      | F=1/2: +1518.12                                        | F=3/2: +1112.00 | F=5/2: +625.25 | F=7/2: +343.65  | F=9/2: -565.98  | F=11/2: -1255.75 |  |  |
| $z^3G_7^o$ | -96.60(20)   | -0(3)      | F=3/2: +1207.38                                        | F=5/2: +985.59  | F=7/2: +627.92 | F=9/2: +193.27  | F=11/2: -338.04 | F=13/2: -966.06  |  |  |
| $y^3D_5^o$ | -14.2(3)     | -0(7)      | F=1/2: +248.06                                         | F=3/2: +146.50  | F=5/2: +106.67 | F=7/2: +13.51   | F=9/2: -92.11   | F=11/2: -213.28  |  |  |

Measured A, B constants; sources in the data file.

| Transition              | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|-------------------------|--------------------|----------|-------------|
| $a^1F_u - {}^3D_0^o$    | 391.544            | 46-48    | -943(3)     |
| $a^1F_u - {}^3D_1^o$    | 391.544            | 50-48    | 884(3)      |
| $a^1F_u - {}^3D_2^o$    | 391.544            | 47-48    | -448(3)     |
| $a^1F_u - {}^3D_3^o$    | 391.544            | 49-48    | 474(3)      |
| $a^1F_7 - {}^3G_6^o$    | 498.312            | 46-48    | -716.2(4)   |
| $a^1F_7 - {}^3G_7^o$    | 498.312            | 50-48    | 671.6(21)   |
| $a^1F_7 - {}^3G_8^o$    | 498.312            | 47-48    | -346(3)     |
| $a^1F_7 - {}^3G_9^o$    | 498.312            | 49-48    | 355(3)      |
| $a^1F_7 - {}^3G_{10}^o$ | 499.246            | 46-48    | -706.7(9)   |
| $a^1F_7 - {}^3G_{11}^o$ | 500.090            | 50-48    | 660.1(6)    |
| $a^1F_7 - {}^3G_{12}^o$ | 500.090            | 46-48    | -703.2(12)  |
| $a^1F_7 - {}^3G_{13}^o$ | 500.090            | 49-48    | 660.1(6)    |
| $a^1F_7 - {}^3G_{14}^o$ | 500.861            | 46-48    | -691.3(23)  |
| $a^1F_7 - {}^3G_{15}^o$ | 500.861            | 50-48    | 652.0(5)    |
| $a^1F_7 - {}^3G_{16}^o$ | 501.567            | 46-48    | -682.7(3)   |
| $a^1F_7 - {}^3G_{17}^o$ | 501.567            | 50-48    | 642.1(4)    |
| $a^1F_7 - {}^3G_{18}^o$ | 501.756            | 46-48    | -714.0(6)   |
| $a^1F_7 - {}^3G_{19}^o$ | 501.756            | 50-48    | 682.9(6)    |
| $a^1F_8 - {}^3G_6^o$    | 502.143            | 46-48    | -702(5)     |
| $a^1F_8 - {}^3G_7^o$    | 502.143            | 50-48    | 670.6(11)   |
| $a^1F_8 - {}^3G_8^o$    | 502.427            | 46-48    | -690.1(5)   |
| $a^1F_8 - {}^3G_9^o$    | 502.427            | 50-48    | 660.29(28)  |

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$